

ONLINE FOOD ORDERING SYSTEM

Database Management System



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Course	DataBase
Semester	CS4-1
Language	Oracle
GitHub	https://github.com/harishabib228/Online-Food-Delivery-System

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1. Introduction

The Online Food Ordering System is a database application that allows customers to order food from restaurants using an online platform. This system manages customer data, restaurant menus, orders, and payments efficiently.

This project applies database concepts such as ERD, normalization, constraints, and advanced SQL queries. It provides a real-world solution for managing food delivery services digitally.

2. Entity Relationship Diagram (ERD)

The ERD represents the structure of the database and shows how different entities are related to each other.

2.1 Entities and Attributes

Customer

- Customer_ID (PK)
- Name
- Email
- Phone

Restaurant

- Restaurant_ID (PK)
- Name
- Location

Menu

- Item_ID (PK)
- Restaurant_ID (FK)
- Item_Name
- Price

Orders

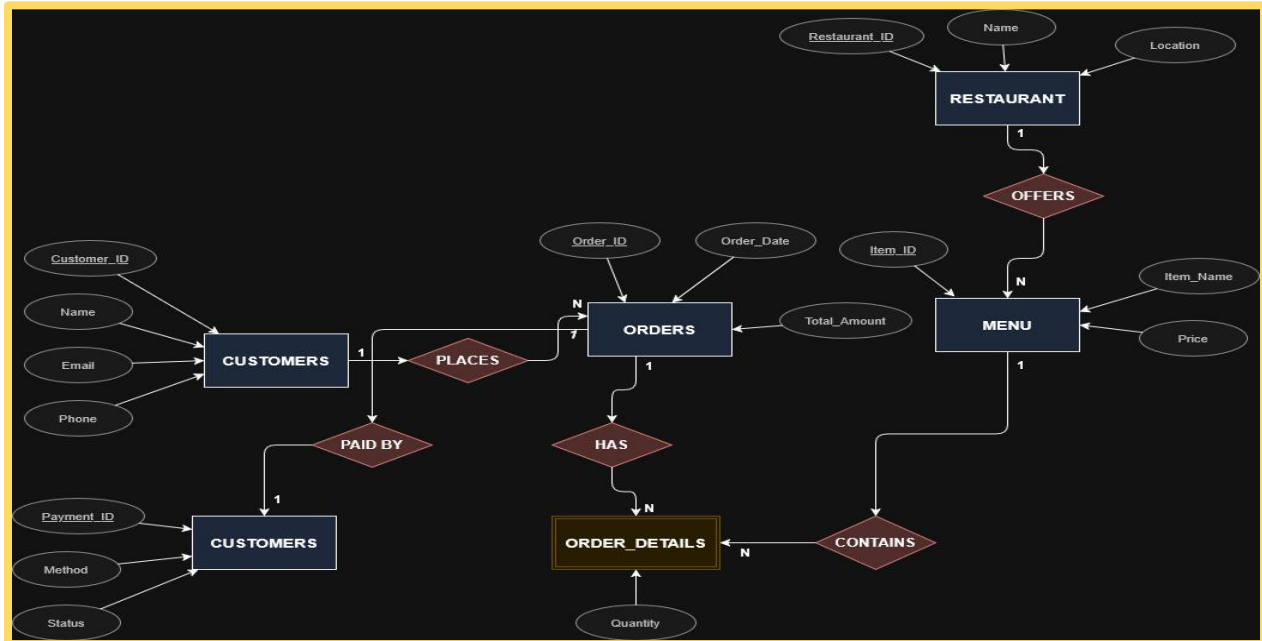
- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount

Order_Details

- Order_ID (PK, FK)
- Item_ID (PK, FK)
- Quantity

Payment

- Payment_ID (PK)
- Order_ID (FK)
- Method
- Status



2.2 Relationships

- One Customer can place many Orders (1:M)
- One Restaurant can have many Menu items (1:M)
- One Order can contain multiple items (1:M via Order_Details)
- One Order has one Payment (1:1)

3. Core Schema Objects

3.1 Tables & Columns

The database consists of the following tables:

- Customer
- Restaurant
- Menu
- Orders
- Order_Details
- Payment

Each table contains relevant fields to store structured data.

3.2 Keys

Primary Keys:

- Customer_ID
- Restaurant_ID
- Item_ID
- Order_ID
- Payment_ID
- (Order_ID, Item_ID) in Order_Details

Foreign Keys:

- Restaurant_ID → Menu
- Customer_ID → Orders
- Order_ID → Order_Details & Payment
- Item_ID → Order_Details

3.3 Constraints

- **PRIMARY KEY** → Ensures uniqueness
- **FOREIGN KEY** → Maintains relationships
- **NOT NULL** → Required fields
- **UNIQUE** → Email in Customer
- **CHECK** → Payment status ('Paid', 'Pending')

3.4 Normalization (up to 3NF)

1NF (First Normal Form):

- All attributes contain atomic values
- No repeating groups

2NF (Second Normal Form):

- All non-key attributes fully depend on primary key

3NF (Third Normal Form):

- No transitive dependency
- Data redundancy removed

Result: Efficient and consistent database design

3.5 Relationships

- Customer → Orders (1:M)
- Restaurant → Menu (1:M)
- Orders → Order_Details (1:M)
- Menu → Order_Details (1:M)

- Orders → Payment (1:1)

3.6 Views

Customer_Orders View:

- Displays customer name, order ID, and total amount
- Helps in generating reports easily

4. Queries

4.1 Join Query

```
SELECT C.Name, O.Order_ID  
FROM Customer C  
JOIN Orders O ON C.Customer_ID = O.Customer_ID;
```

Shows customer names with their orders

4.2 Subquery

```
SELECT Name FROM Customer  
WHERE Customer_ID IN (SELECT Customer_ID FROM Orders);
```

Shows customers who have placed orders

4.3 Aggregation Function

```
SELECT SUM(Total_Amount) FROM Orders;
```

Calculates total sales

5. Complex Computing Problem

Problem: Find the customer with highest total spending

```
SELECT Customer_ID, SUM(Total_Amount) AS Total_Spent  
FROM Orders  
GROUP BY Customer_ID  
HAVING SUM(Total_Amount) =  
(  
    SELECT MAX(SUM(Total_Amount))  
    FROM Orders  
    GROUP BY Customer_ID  
);
```

Identifies top customer based on spending

6. Conclusion

The Online Food Ordering System is a well-structured database project that demonstrates key DBMS concepts. The system includes proper ERD design, normalized tables, constraints, and advanced SQL queries. It efficiently manages customers, orders, and payments, making it a practical real-world application.